LABORATORY · Lab · 26 hours

Experiment 7: Design of a PID Controller Using Tuning Methods



Control Systems Laboratory (22EEG53)

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Learning objectives

- Design a PID controller using Ziegler–Nichols tuning
- Simulate the closed-loop step response
- Compare P, PI and PID responses
- Fine-tune the gains

Syllabus context

Laboratory component of 22EEG53 Control Systems (0:0:2 hours per week). CIE: record and observation 25, viva-voce 5, lab test 20 marks.

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Aim and requirements

Aim: to design a PID controller using tuning methods in MATLAB.

Software: MATLAB with the Control System Toolbox (Simulink where required).

Hardware: Personal computer.

Plant: $G(s) = 1/[s(s + 1)(s + 2)]$; $K_{cr} = 6$, $\omega_{cr} = 1.414$ rad/s, $P_{cr} = 4.44$ s.

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Ziegler–Nichols settings

Controller	Kp	Ti	Td
P	0.5 Kcr	∞	0
PI	0.45 Kcr	Pcr/1.2	0
PID	0.6 Kcr	0.5 Pcr	0.125 Pcr

PID values here: Kp = 3.6, Ti = 2.22 s, Td = 0.555 s.

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Procedure

1

Find Kcr

Use margin(G) or Routh: K = 6 gives sustained oscillation.

2

Compute settings

Apply the ZN table.

3

Build controller

C = pid(Kp, Ki, Kd).

4

Simulate

step(feedback(C*G, 1)); compare with P, PI.

5

Fine-tune

Adjust to reduce overshoot.

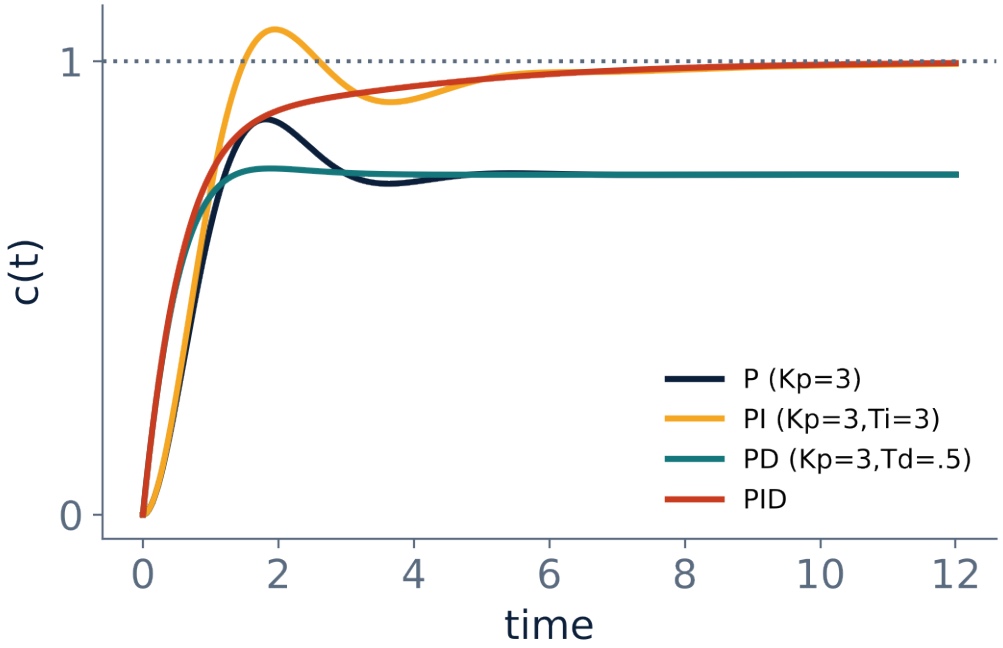
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MATLAB program

```
G = tf(1, conv([1 0], conv([1 1], [1 2])));
[Gm, ~, wcr] = margin(G);  Kcr = Gm;  Pcr = 2*pi/wcr;
Kp = 0.6*Kcr;  Ti = 0.5*Pcr;  Td = 0.125*Pcr;
C_pid = pid(Kp, Kp/Ti, Kp*Td);
C_pi  = pid(0.45*Kcr, 0.45*Kcr/(Pcr/1.2));
C_p   = pid(0.5*Kcr);
t = 0:0.05:40;  hold on
step(feedback(C_p*G,1), t), step(feedback(C_pi*G,1), t),
step(feedback(C_pid*G,1), t)
legend('P','PI','PID'); grid on
```

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Typical responses (illustration)



- PID gives lowest steady-state error with acceptable overshoot.
- P alone leaves error for type-0 plants.
- Fine-tune Kd to damp oscillation.

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Observation table

Controller	Kp	Ki	Kd	Mp (%)	ts (s)	ess
P						
PI						
PID	3.6	1.62	2.0			

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Key takeaways

- Find Kcr and Pcr, then use ZN formulas
- PID: Kp = 0.6Kcr, Ti = 0.5Pcr, Td = 0.125Pcr
- I removes the steady-state error; D adds damping
- Fine-tune after ZN to reduce overshoot

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Practice questions

Q1. List the ZN settings for P, PI and PID.

Q2. Find Kcr and Pcr for K/[s(s+2)(s+4)].

Q3. Effect of Kd on overshoot?

Q4. Write the MATLAB command to create a PID controller.